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Homogeneity of Rainfall and Temperature Series in Bangladesh

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Abstract: An error free data set is prerequisite in climatic study. The objective of this study is to test homogeneity of the climatic data with a view to determining its suitability in exercising climate change research. Homogeneity of monthly temperature and rainfall series of 32 BMD (Bangladesh Meteorological Department) stations collected from Bangladesh Agricultural Research Council were studied. The different time scale was used for different stations based on the availability of data. Standard Normal Homogeneity Test (SHNT), Buishand range test, Pettitt test and Von Neumann ratio test were performed using monthly data. The results of these tests were summarised into three classes: 'useful', 'doubtful' and 'suspect'. A qualitative interpretation and some recommendations to use these labelled series are also proposed. It was found that 94% rainfall series and 59 % of temperature series could be labelled as 'useful'. There are 3% rainfall series and 19% temperature series classified as 'doubtful'. Rest 3% rainfall series and 22% temperature series can be classified as 'suspect'.

Keywords: Homogeneity test, climate data